Astroshop deployment topology

How the services are laid out across nodes and networks. This is a two-column document with a tight gutter; if your extraction interleaves the columns, the reading order score will show it.

accounting (C#)

Consumes the orders topic from Kafka and writes ledger entries. Fully asynchronous. Tier 2, owned by data. Calls kafka. Scrape target on the metrics network; exports spans to the collector over OTLP.

ad (Java)

Serves a contextual ad on the product page. Leaf service under frontend. Tier 1, owned by catalog. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

astronomy-db (SQL)

Relational store behind product-catalog. Tier 1, owned by platform-eng. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

cart (C#)

Holds the customer's cart. Backed by valkey-cart (Redis-compatible). Health-checked with grpc_health_probe; a failing check marks it unhealthy but never restarts it. Tier 0, owned by checkout. Calls valkey-cart. Co-located with its datastore. Scrape target on the metrics network; exports spans to the collector over OTLP.

checkout (Go)

The order pipeline. Gets the cart, prices each item (product-catalog + currency), quotes shipping, charges payment, books shipping, empties the cart, emails a confirmation, then publishes the order to Kafka. Tier 0, owned by checkout. Calls cart, currency, email, payment, product-catalog, shipping. Scrape target on the metrics network; exports spans to the collector over OTLP.

currency (C++)

Converts prices between currencies. Called by frontend and checkout. Pure compute, no dependencies. Tier 1, owned by checkout. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

email (Ruby)

Sends the order-confirmation email. Called by checkout after the order is committed, best-effort. Tier 2, owned by checkout. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

flagd (Go)

Feature-flag daemon. Reads demo.flagd.json and serves flag evaluations. The fault-injection control plane. Tier 1, owned by platform-eng. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

fraud-detection (Kotlin)

Consumes the orders topic and scores each order for fraud. Asynchronous, does not gate checkout. Tier 2, owned by data. Calls kafka. Scrape target on the metrics network; exports spans to the collector over OTLP.

frontend (TypeScript)

Serves the web storefront and the JSON API the browser calls. Fans out to every product path. Tier 0, owned by storefront. Calls ad, cart, checkout, currency, product-catalog, quote, recommendation, shipping, image-provider. Scrape target on the metrics network; exports spans to the collector over OTLP.

frontend-proxy (C++)

Edge proxy. All external traffic and the flagd-ui and telemetry-docs routes pass through it. Tier 0, owned by storefront. Calls frontend. Scrape target on the metrics network; exports spans to the collector over OTLP.

image-provider (Nginx)

Static image host for product photos. Served through frontend-proxy. Tier 1, owned by storefront. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

kafka (Java)

The order event bus. checkout produces to it; accounting and fraud-detection consume. Tier 1, owned by data. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

otel-collector (Go)

Receives OTLP from every service, batches, and exports to Prometheus and Jaeger. Excluded from the health rollup by design. Tier 1, owned by platform-eng. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

payment (JavaScript)

Charges the card. Pure compute stub, no real PSP. The last synchronous call before the order is confirmed. Tier 0, owned by checkout. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

product-catalog (Go)

The product database. Shared dependency of frontend and recommendation. Backed by astronomy-db. Tier 0, owned by catalog. Calls astronomy-db. Co-located with its datastore. Scrape target on the metrics network; exports spans to the collector over OTLP.

quote (PHP)

Computes a shipping quote (weight/size -> price). Called by shipping. Tier 1, owned by catalog. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.

recommendation (Python)

Returns "you may also like" products. Calls product-catalog. Has an in-process cache. Tier 1, owned by catalog. Calls product-catalog. Scrape target on the metrics network; exports spans to the collector over OTLP.

shipping (Rust)

Quotes a shipping cost and books the shipment. Called by checkout. Tier 1, owned by checkout. Calls quote. Scrape target on the metrics network; exports spans to the collector over OTLP.

valkey-cart (C)

In-memory store behind cart. Redis wire protocol. Tier 1, owned by platform-eng. No outbound service calls. Scrape target on the metrics network; exports spans to the collector over OTLP.